SIEM & SOAR Architecture

The diagram covers the full 7-layer SIEM + SOAR architecture with clear visual separation between both systems:

SIEM Layers (purple)

- **L1 – Telemetry Ingestion**: Endpoint/OS logs, network/perimeter, cloud/SaaS, app/DB, identity/IAM, and container/K8s sources
- **L2 – Collection & Normalization**: Kafka event bus, log forwarders, ECS/OCSF normalization, real-time enrichment (GeoIP, asset context), and deduplication
- **L3A – SIEM Core**: CEP-based real-time correlation, SIGMA/YARA-L rule engine, ML/UEBA behavioral analytics, and risk-based alert scoring

SOAR Layers (blue)

- **L3B – SOAR Core**: Visual DAG playbook orchestrator, 800+ integration broker, sandboxed action executor, and case/incident state machine
- **L6 – Response Actions**: Endpoint containment (CrowdStrike/S1), network blocking, identity response (AD/Okta), cloud remediation, and ticketing

Shared Layers (green)

- **L4 – Threat Intelligence**: STIX/TAXII feeds, sub-millisecond IoC matching, ATT&CK TTP mapping, and threat hunting workflows
- **L5 – Storage**: Hot (Elasticsearch), cold (Parquet/S3), relational case DB, Redis cache, and Prometheus metrics
- **L7 – Analyst UI**: SOC dashboard, log search, incident workspace, visual playbook designer, and compliance reporting

Security Information & Event Management · Security Orchestration, Automation & Response

LAYER 1 -Telemetry & Data Source IngestionSIEM — INPUT

Endpoint & OS Logs

Windows Event Logs (WEL), Linux syslog/auditd, macOS unified logs. Agent-based (Wazuh/Filebeat) and agentless WMI collection.

WEL/AUDITD/WAZUH/FILEBEAT

Network & Perimeter

Firewall logs, NetFlow/IPFIX, DNS query logs, proxy/web logs, IDS/IPS alerts (Suricata, Snort), Zeek conn logs.

NETFLOW/SURICATA/ZEEK/PCAP

Cloud & SaaS

AWS CloudTrail, Azure Monitor, GCP Audit Logs. SaaS connectors for Office 365, Google Workspace, Salesforce, Okta.

CLOUDTRAIL/AZURE MONITOR/OKTA

App & DB Logs

Application error logs, DB audit trails (Oracle, MSSQL, PostgreSQL), API gateway access logs, WAF events.

WAF/DB AUDIT/APP LOGS

Identity & IAM

Active Directory / LDAP events, PAM session logs, MFA challenge/response, SCIM provisioning events, VPN auth logs.

AD-LDAP/PAM/MFA/VPN

Container & K8s

Kubernetes audit logs, container runtime events (Falco), image scan results, pod security policy violations.

FALCO/K8S AUDIT/OCI

↓SYSLOG · KAFKA STREAMS · BEATS · API PULL · UDP/TCP 514↓

LAYER 2 - Collection, Transport & Normalization

HIGH-THROUGHPUT EVENT PIPELINE · EPS-AWARE BACK-PRESSURE · SCHEMA-ON-WRITE

Log Collector / Forwarder

Distributed forwarder fleet (Logstash, Fluentd, Vector). TLS-encrypted transport, compression, at-least-once delivery guarantees.

LOGSTASH/FLUENTD/VECTOR

Message Broker

Apache Kafka as central event bus. Topic-per-source-type, consumer groups for parallel processing, 7-day retention for replay.

KAFKA/CONSUMER GROUPSAVRO

Normalization Engine

ECS / OCSF field mapping. Vendor-specific parsers (500+ built-in). Grok patterns, regex extraction, CEF/LEEF translation.

ECS/OCSF/CEF/GROK

Enrichment Pipeline

Real-time GeoIP tagging, hostname resolution, asset context lookup, user identity resolution from LDAP/IdP, CIDR classification.

GEOIP/DNS RESOLVE/LDAP LOOKUP

Dedup & Suppression

Hash-based deduplication window. Suppression rules for known noisy sources. Rate-limiting per source to protect indexing capacity.

BLOOM FILTERT/TL WINDOW

↓NORMALIZED ENRICHED EVENTS → INDEX SHARDS + CORRELATION STREAM↓

LAYER 3A – SIEM Core Engine Detection

Real-Time Correlation Engine

CEP-based event stream processing. Sliding & tumbling window rules. Multi-event pattern matching across millions of EPS. Stateful rule evaluation.

CEP/FLINK/SLIDING WINDOW

Rule & Sigma Engine

SIGMA rule compilation, YARA-L rules, custom DSL for correlation. Built-in rule library (ATT&CK-mapped). Rule versioning & A/B testing.

SIGMA/YARA-L/ATT&CK

ML / UBA Engine

User & Entity Behavior Analytics. Baseline profiling per user/host, peer-group comparison, anomaly scoring, insider threat models.

UEBA/ISOLATION FOREST/LSTM

Alert Triage & Scoring

Risk-based alert scoring combining rule confidence, asset criticality, threat intel hit, and UEBA score. Alert fatigue suppression via ML.

RISK SCORE/PRIORITY QUEUE

L3B - SOAR Core Engine ORCHESTRATION

Playbook Orchestrator

Visual DAG-based playbook builder. Conditional branching, parallel execution, human-in-the-loop gates. Version control & audit trail for all runs.

DAG/AIRFLOW/BPMN

Integration Broker

800+ pre-built integrations: EDR, firewalls, ticketing, identity, cloud APIs. OAuth2 credential vault, retry logic, rate-limit awareness.

REST/GRAPHQL/GRPC

Action Executor

Sandboxed action execution environment. Rollback-capable containment actions. Dry-run mode, change approval workflows, blast-radius estimation.

SANDBOXED/IDEMPOTENT

Case & Incident Manager

Incident lifecycle state machine: New → Investigating → Contained → Resolved. Timeline, evidence locker, analyst notes, SLA clock, escalation rules.

STATE MACHINE/SLA/EVIDENCE

↓ENRICHED ALERTS · INCIDENT RECORDS · INVESTIGATION CONTEXT↓

LAYER 4 - Threat Intelligence Platform (TIP)**INTEL**

Intel Feed Aggregator

Commercial (Recorded Future, Mandiant), open-source (MISP, OpenCTI), ISAC sharing circles, government feeds (CISA AIS). TAXII/STIX 2.1 ingestion.

STIX 2.1/TAXII/MISP

IoC Matching Engine

Sub-millisecond IoC lookups against live event stream. IP/domain/hash/URL matching. Bloom filter + exact-match hybrid. Confidence-weighted hits.

BLOOM FILTER/REDIS BITMAP/TTL DECAY

ATT&CK Mapper

Automatic MITRE ATT&CK TTP tagging on alerts. Tactic/technique/sub-technique resolution. Navigator heatmap generation for coverage gap analysis.

ATT&CK V15/NAVIGATOR

Intel Lifecycle Manager

TLP-aware sharing controls, IoC aging and TTL expiry, deduplication across feeds, confidence scoring, feedback loop from incident outcomes.

TLP/CONFIDENCE SCORE/TTL

Threat Hunting Module

Hypothesis-driven hunt workflows, retroactive IoC sweep across historical data, pivot investigation from any artifact, hunt pack library.

RETROACTIVE/HUNT PACKS/PIVOT

↓IOC HITS · TTP TAGS · THREAT SCORES → ALERT ENRICHMENT + PLAYBOOK TRIGGER↓

LAYER 5 - Data Storage, Indexing & Retention **PERSISTENCE**

Hot Search Index

Elasticsearch / OpenSearch cluster for active 90-day window. Full-text and field-level search, aggregations, real-time indexing at 200k+ EPS. Cross-cluster search for federated queries across shards.

ELASTICSEARCH/OPENSEARCH/ILM/CROSS-CLUSTER

Cold / Archive Tier

S3-compatible object storage for 7-year log archive. Compressed Parquet + columnar format. Query-on-demand via Athena/Trino.

PARQUET/S3/ATHENA

Case & Incident DB

Relational store for incident records, playbook run logs, action audit trail, analyst notes, evidence artifacts and attachments.

POSTGRESQL/IMMUTABLE LOG

Real-time Cache

Redis for alert queues, live dashboard metrics, IoC hot-cache, session data, rate-limit counters, and playbook execution state.

REDIS CLUSTER/PUB-SUB

Metrics Store

Time-series metrics (Prometheus + VictoriaMetrics) for pipeline health, EPS rates, rule hit counts, MTTR/MTTD KPIs, and SLA tracking.

PROMETHEUS/VICTORIAMETRICS

↓QUERY RESULTS · INCIDENT DATA · METRICS → API + DASHBOARDS↓

LAYER 6 - Automated Response & Containment Actions **SOAR — RESPONSE**

Endpoint Containment

EDR-driven host isolation (CrowdStrike, SentinelOne, Defender). Process kill, memory dump trigger, forensic evidence preservation before quarantine.

CROWDSTRIKE/S1/DEFENDER

Network Blocking

Dynamic firewall ACL push (Palo Alto, Cisco ASA, NSGs), BGP null-route for IP blocks, DNS sinkhole for malicious domains, proxy category override.

PALO ALTO/BGP NULL-ROUTE/DNS SINK

Identity Response

Force MFA re-auth, revoke OAuth tokens, disable AD account, reset credentials, remove group memberships, trigger PAM session recording.

AD-AZURE AD/OKTA/CYBERARK

Ticketing & Notification

Auto-create Jira/ServiceNow incidents with evidence attached. PagerDuty escalation, Slack/Teams alerts, email notifications with context-rich summaries.

JIRA/SERVICENOW/PAGERDUTY

Cloud Remediation

Revoke cloud IAM policies, snapshot & terminate compromised EC2/VMs, modify Security Group rules, rotate access keys, quarantine S3 buckets.

AWS IAM/AZURE RBAC/GCP

↓PROCESSED DATA · API RESPONSES · LIVE EVENTS → UI↓

LAYER 7 - Analyst UI, Dashboards & Reporting **PRESENTATION**

SOC Dashboard

Real-time alert queue, EPS heatmap, open incident board, MTTD/MTTR KPIs, analyst workload, threat map globe view.

REACT/WEBSOCKET/D3.JS

Log Search & Investigation

Elasticsearch-powered full-text log search. KQL/Lucene query bar, field picker, timeline view, saved searches, pivot from any field value.

KQL/LUCENE/PIVOT

Incident Workspace

Unified case view: timeline, evidence locker, analyst collaboration, playbook run status, linked alerts, ATT&CK kill-chain visualization.

TIMELINE/COLLABORATION/ATT&CK

Playbook Designer

Visual drag-and-drop playbook builder. Action library browser, test/simulation mode, version diff, deployment approval workflow.

NO-CODE/VISUAL DAG/SIM MODE

Reporting & Metrics

Executive reports, compliance-ready evidence export, MTTR/MTTD trend charts, coverage heatmaps. Scheduled PDF delivery to stakeholders.

PDF/SCHEDULED/COMPLIANCE

Cross-Cutting Concerns

High Availability

Active-active multi-region. Kafka replication factor 3, ES replica shards, Redis Sentinel. RTO <5 min.

Encryption

TLS 1.3 in-transit, AES-256 at rest, KMIP key management, HSM-backed master keys for regulated data.

Multi-tenancy

Tenant-scoped data partitioning, RBAC namespace isolation, per-tenant Kafka topics and ES indices.

Observability

Distributed tracing (Jaeger/Tempo), Prometheus metrics, structured JSON logs, Grafana dashboards for pipeline health.

Immutable Audit Log

All analyst actions, playbook runs, config changes logged to append-only store. SHA-256 chain. 7-year retention.

Change Control

Rule & playbook changes require peer review, staging validation, and canary rollout before production push.

Integrated EDR / XDR

CrowdStrike Falcon

Host isolation, process tree, RTR remote shell

SentinelOne

Automated remediation, threat storyline API

Microsoft Defender XDR

M365 signal, identity + endpoint unified

Palo Alto Cortex

XDR + XSOAR native integration

Carbon Black

VMware env coverage, enterprise telemetry

SOAR Playbook Types

Phishing Response

Extract IoCs → block URLs → quarantine email → notify user → reset creds

Ransomware Containment

Isolate host → snapshot disk → kill process → notify IR team → open P1 ticket

Insider Threat

Corroborate UEBA score → pull HR context → disable account (pending review) → preserve evidence

Brute Force

Threshold check → GeoIP verify → force MFA → block IP at WAF → alert analyst

Cloud Credential Leak

Revoke key → audit usage → notify owner → generate new key → update secrets vault

Compliance & Frameworks

MITRE ATT&CK v15

Full TTP coverage mapping for all detection rules

NIST SP 800-61

IR lifecycle: Prepare → Detect → Contain → Eradicate → Recover

SOC 2 / ISO 27001

Audit trail, access logging, retention policies

PCI-DSS v4.0

Log integrity, 12-month retention, anomaly detection reqs

GDPR / HIPAA

PII masking in logs, breach notification playbooks, DPA workflows

Deployment Topology

Kubernetes

Helm-deployed microservices, HPA scaling on EPS load, namespace-per-tenant

Multi-Region Active/Active

AWS us-east-1 + eu-west-1 primary nodes, GCP failover DR site

Air-Gapped Option

On-premise deployment for classified / government environments